

Answer: D

- 29** When R and X are in series, $I = 0.3 \text{ A}$ (given)

From the graph, $V_R = 1.0 \text{ V}$ and $V_X = 1.5 \text{ V}$.

Hence e.m.f. = $(1.0 + 1.5)\text{V} = 2.5 \text{ V}$

When R and X are in parallel, $V_R = 2.5 \text{ V}$ and $V_X = 2.5 \text{ V}$.

From the graph, when $V = 2.5 \text{ V}$, $I_R = 0.5 \text{ A}$ and $I_X = 0.5 \text{ A}$

Current in the cell = $(0.5 + 0.5)\text{A} = 1.0 \text{ A}$

Answer: D

- 30** Applying the right-hand grip rule, magnetic fields due to all 3 coils are out of the page in